

1. Introduction

1.1. Assignment Specifications

This project simulates a digital communication downlink meeting the requirement of transmitting >2048 bits within a bandwidth <20% of the carrier frequency under AWGN conditions (6, 18, 24 dB). Beyond core requirements, the architecture approximates 3GPP LTE Release 8 logic to analyse performance in urban environments. Key "creative additions" include Adaptive Modulation and Coding (AMC) (switching between QPSK and 16-QAM), Convolutional Coding with Viterbi decoding, Stop-and-Wait ARQ, and Jakes' Model Rayleigh fading to simulate Non-Line-of-Sight (NLOS) conditions.

1.2. Generative 'AI' Statement:

Generative AI (Gemini 3 Pro) was utilized exclusively for code debugging, commenting, and proofreading. All simulation logic and parameter definitions were derived independently.

2. Simulation Method and Parameters

2.1. Choices and Justification basis

The simulation architecture is an attempt to approximate the physical and Medium Access Control (MAC) layers of an LTE transmitter in a downlink scenario. In trying to align with 3GPP R8 in an urban environment, the system uses adaptive modulation and coding to maximise performance at various channel states. The choice of Rayleigh fading utilizing Jakes' Model is justified to represent the non line of sight multipath propagation that is characteristic of an urban environment (note here that the HATA model is not used as the simulation parameters do not match its requirements; Frequency < 150Mhz)

2.2. Methods

2.2.1. Packetization and Data Framing

The input binary stream is segmented into fixed length payloads (64bits in this case) and encapsulated in a pseudo-MAC PDU style 40 bit prepended header. This block contains an 8 bit sequence number, for ARQ ordering and stitching data together and a 32 bit predefined known string for checking representative error rate to determine modulation and ARQ logic.

2.2.2. Dynamic modulation

The system leverages the BER of the known string in the previous transmission loop applied via a rudimentary uplink loop (simulated). The modulation method is QPSK by default and is switched automatically, if the BER < 0.1%, using the higher order modulation, 16-QAM, in the next transmission to increase efficiency. Subsequent failures mean it falls back on to QPSK.

2.2.3. Quadrant Phase Shift Keying (QPSK)

Used as the default modulation method, the bits are mapped onto the complex symbol $d = I + jQ$ with $I, Q \in \{-1, 1\}$ and I are odd bits and Q are even bits. Providing a spectral efficiency of 2 bits/symbol with high tolerance to noise and poor channel conditions.

2.2.4. 16 Quadrant Amplitude Modulation (16-QAM)

Used, when possible, for higher throughput communication, the modulator maps 4 bit touples to the same complex symbol but with $I, Q \in \{-3, -1, 1, 3\}$. Providing a spectral efficiency of 4 bits/symbol with lower tolerance to noise and poor channel conditions. To maintain the same power as the QPSK transmission, the modulation is normalised by a factor $\sqrt{10}$.

2.2.5. Convolutional Encoding and decoding (Viterbi decoding)

Encoding: A $R = 1/2$, (5, 7), constraint length $K = 3$ non recursive convolutional encoder is implemented. The choice of convolution is not only based on the LTE spec but also because it is highly effective in urban environments.

Decoding: The receiver function uses a Viterbi decoding algorithm with hard decision logic. It computes the hamming distance between received pairs and the expected branch and forms a trellis selecting the survivor path with the lowest total error metric. (Maximum likelihood sequence estimation)

2.2.6. Root Raised Cosine filter and filter matching

An RRC filter specifically used to maximise bandwidth use, and minimise interference between symbols. A matched pair is implemented by having the same filtering in the transmitter and reciver, resulting in a full raised cosine response. The roll off factor (somewhat arbitrarily) is set to $\alpha = 0.35$.

2.2.7. Channel noise, path loss, and fading

Free space path loss is implemented using a carrier frequency and 1km. Multipath fading is emulated on a smaller scale using Jakes' Model, that utilizes sums of $N = 50$ sinusoids. A doppler shift is applied to model a moving target with various speeds. Using a power calculation an additive gaussian white noise is added to the signal at various levels

2.2.8. Stop and Wait Automatic Repeat Request

Being a Robust Error control method, the receiver uses the known string to estimate the state of the bits package received and if errors are detected, an automatic request for retransmission is placed. This is repeated until the max re-transmission count, which is set to 16, is reached.

2.3. Analysis methods

Single Transmission: Limited utility, develop the algorithm, visually check output data, generate examples.

Monte Carlo Simulation: Due to the statistical variations from run to run, the analysed outputs are based on averaged metrics over 100 transmissions for each scenario.

Outputs: Bit Error Rate: *errors: total bits*, representative of overall performance; efficiency:

total packets: number of transmissions, representative of loss rate; transmission time, representative of simulation method used.

2.4. Simulation Parameters

Parameter	Carrier Frequency	Bandwidth	Velocity	Modulation	Coding Rate	Roll-off (α)
Value	233.5 kHz	~3100 kHz	5.5 km/h	QPSK, 16-QAM	1/2	0.35
Note	Scaled down for simulation efficiency.	Implied by Baud/Filter settings.	Representative of Urban Pedestrian UE.	LTE Rel-8 Standard Modulation schemes.	Standard convolutional coding rate.	Standard pulse shaping factor.

3. Results and analysis

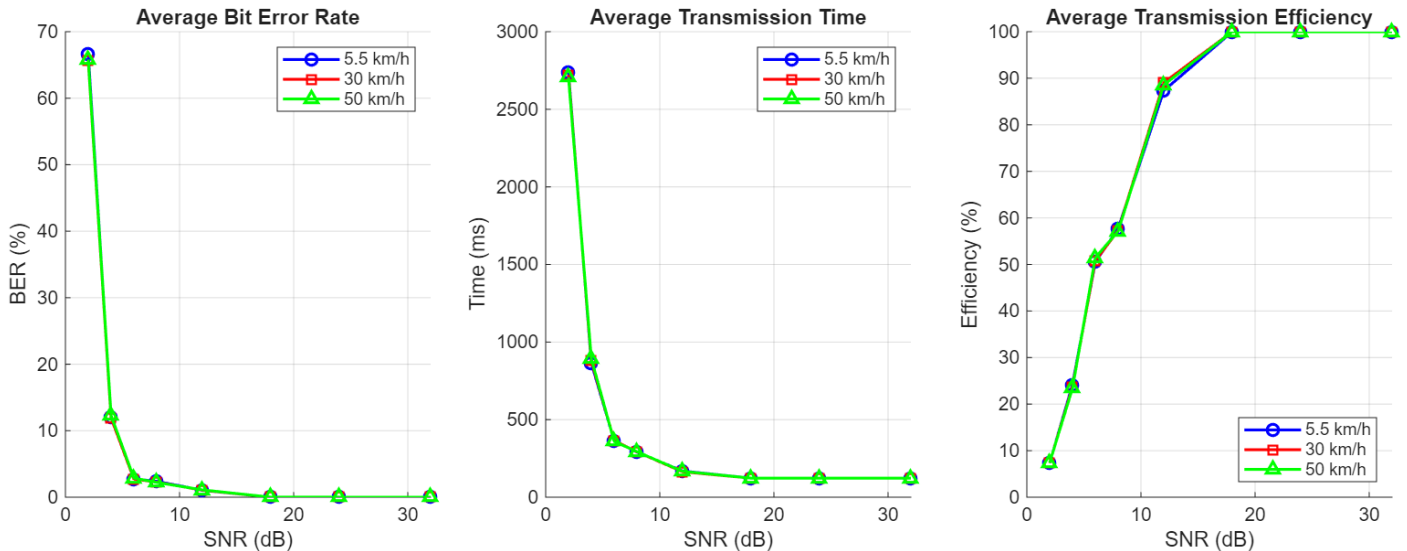
3.1. Single Simulation Outputs (representative examples)

Original: '[20080312T1400Z][frm]00923312511101[\frm][to]00447878503732[\to][MSG]National

Semiconductor fired 1725 people, Bernie Madoff plead guilty, US drone strike kills 12 people in Pakistan, A Sikorsky 92A crashed in Canada and killed 1, ISS astronauts shelterd from space debris in the Russian escape pod[MSG]'

S N R	6	18	24
O u t p u t	[20080312T0400Z][frm]00923312511101[\frm@óto]004478785037SOZ [to][PG]Natioal Semiconductar fired 1725 pe??1, Bernie Mkdoff plead guilty, US drkndôsuðike @ lls 12 peopl3 in Pakistan, A Si}orsky >2A crashed in Canada and killed 1, ISS astronauts shelterd from space debçis in the Sussian escape pz\$[MSG]	[20080312T1400Z][frm]00923312511101[\frm][to]00447878503732[[to][MSG]National Semiconductor fired 1725 people, Bernie Madoff plead guilty, US drone strike kills 12 people in Pakistan, A Sikorsky 92A crashed in Canada and killed 1, ISS astronauts shelterd from space debris in the Russian escape pod[MSG]	[20080312T1400Z][frm]00923312511101[\frm][to]00447878503732[[to][MSG]National Semiconductor fired 1725 people, Bernie Madoff plead guilty, US drone strike kills 12 people in Pakistan, A Sikorsky 92A crashed in Canada and killed 1, ISS astronauts shelterd from space debris in the Russian escape pod[MSG]
B E R	3.125%	0%	0%

3.2. Monte Carlo Simulation



3.3. Analysis

The system exhibits the expected 'waterfall' BER characteristic of coded systems. Low SNR (2-4), effective loss of signal with a ~70% BER. Medium SNR (6-12), some signal is maintained with BER ~2.7%. High SNR (>18) a perfect signal is delivered to the receiver ~0% BER. While there is a very minimal impact of the doppler shift is visible, the impact would be more drastic with higher frequency. The interaction between ARQ and AMC resulted in a 3x reduction in transmission time as SNR improved (364ms at 6 dB vs. 122.8ms at ≥18 dB). Efficiency rose from ~50% to 100%, confirming that while ARQ ensures data integrity by discarding corrupted packets in low SNR, the switch to 16-QAM in high SNR regimes maximizes spectral efficiency. From the simulation we can see that an approximate threshold for the Zero error regime is just above and SNR of 18, which is as expected slightly higher than the theoretical minimum threshold given the fading added. The implementation of a hard decoding algorithm in the Viterbi Algorithm is also a simplification and area for improvement. Implementing soft decision making would likely lower the thresholds further and prevent decision trees from diverging under certain conditions.

4. Conclusion

The project demonstrates a functional, representative model of a modern digital communication link. It highlights the trade-offs inherent in wireless design: the balance between throughput (16-QAM) and robustness (QPSK), and the cost of reliability (ARQ retransmissions) on latency. The simulation successfully achieved all the core requirements. A message exceeding 2048 bits was successfully packetized, modulated, and recovered. The spectral constraints were satisfied using Root Raised Cosine (RRC) filtering ($\alpha=0.35$), and the system was rigorously tested against the required SNR benchmarks (6, 18, 24 dB), providing a clear characterization of performance in noise. The implementation of Adaptive Modulation and Coding (AMC) was validated by the 3x reduction in transmission time at high SNR. The Convolutional Coding ($K=3$) provided visible coding gain in the transition region (6-12dB). The ARQ system functioned correctly to ensure data integrity, prioritizing accuracy over latency.

Appendix 1: Code

The code is contained in the python files submitted:

Full_Single.py: simulates a single transmission loop

Monte_Carlo_Sim.py: 100 averaged simulations of the transmission loop

Appendix 2: Monte Carlo Raw results

	Average Bit Error Rate (%)			Average Transmission Time (ms)			Average Transmission Efficiency (%)		
SNR	5.5km/h	30km/h	50km/h	5.5km/h	30km/h	50km/h	5.5km/h	30km/h	50km/h
2	66.62	65.63	65.82	2736.8	2721.6	2706.4	7.33	7.37	7.41
4	11.97	11.93	12.28	865.6	879.1	892.7	24.15	23.72	23.43
6	2.75	2.74	2.87	364.7	364.1	359.9	50.53	50.79	51.44
8	2.39	2.27	2.23	290.6	292.4	293.3	57.68	57.29	57.14
12	1.04	1.06	1.1	168.5	163.9	166.5	87.34	89.05	88.56
18	0.0008	0.0004	0.0004	122.8	122.8	122.8	100	100	100
24	0	0	0	122.8	122.8	122.8	100	100	100
32	0	0	0	122.8	122.8	122.8	100	100	100

Appendix 3: References

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